

ADVANCED

Stage 6

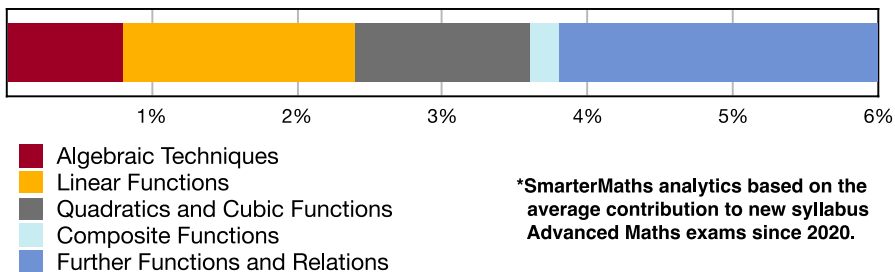
Functions (Adv), F1 Working with Functions (Adv)

Algebraic Techniques (Y11)

Teacher: SHS Maths

Exam Equivalent Time: 81 minutes (based on allocation of 1.5 minutes per mark)

F1 Working With Functions



HISTORICAL CONTRIBUTION

- F1 Working with Functions* is classic low hanging fruit, contributing a healthy 6.0% per exam since the new Advanced syllabus was introduced in 2020.
- We have split the topic into 5 categories for analysis purposes: 1-*Algebraic Techniques* (0.8%), 2-*Linear Functions* (1.6%), 3-*Quadratics and Cubic Functions* (1.2%), 4-*Composite Functions* (0.2%) and 5-*Further Functions and Relations* (2.2%).
- This analysis looks at *Algebraic Techniques*.

HSC ANALYSIS - What to expect and common pitfalls

- Algebraic Techniques* (0.8%) has been a minor contributor to new syllabus exams although it covers crucial skills required throughout later topics.
- Surds have traditionally been most common question type but as yet, have not attracted a dedicated in new syllabus exams. We note that prior to 2020, most past questions involved compound surds in the denominator which are no longer examinable.
- Algebraic Fractions were last examined in the 2021 Advanced paper and deserve attention, particularly as they represent an overlap between the new Advanced and Standard 2 content.
- Rounding questions can require students to know standard decimal place rounding, significant figures (2019) and scientific notation (2015).

Questions

1. Functions, 2ADV F1 2018 HSC 1 MC

What is the value of $7^{-1.3}$ correct to two decimal places?

- A. 0.07
- B. 0.08
- C. -12.54
- D. -12.55

2. Functions, 2ADV F1 2012 HSC 1 MC

What is 4.09784 correct to three significant figures?

- A. 4.09
- B. 4.10
- C. 4.097
- D. 4.098

3. Functions, 2ADV F1 2014 HSC 1 MC

What is the value of $\frac{\pi^2}{6}$, correct to 3 significant figures?

- A. 1.64
- B. 1.65
- C. 1.644
- D. 1.645

4. Functions, 2ADV F1 2015 HSC 1 MC

What is 0.005 233 59 written in scientific notation, correct to 4 significant figures?

- A. 5.2336×10^{-2}
- B. 5.234×10^{-2}
- C. 5.2336×10^{-3}
- D. 5.234×10^{-3}

5. Functions, 2ADV F1 2019 HSC 1 MC

What is the value of π^{10} to two significant figures?

- A. 9.36×10^4
- B. 9.4×10^4
- C. 9.36×10^5
- D. 9.4×10^5

6. Functions, 2ADV F1 2005 HSC 1a

Evaluate $\sqrt{\frac{275.4}{5.2 \times 3.9}}$ correct to two significant figures. *(2 marks)*

7. Functions, 2ADV F1 2007 HSC 1a

Evaluate $\sqrt{\pi^2 + 5}$ correct to two decimal places. *(2 marks)*

8. Functions, 2ADV F1 SM-Bank 50

Rationalise the denominator of $\frac{1}{4\sqrt{3}}$. *(2 marks)*

9. Functions, 2ADV F1 SM-Bank 53

i. If $\frac{1}{\sqrt[3]{7 + \pi}} = (7 + \pi)^x$, find x . *(1 mark)*

ii. Calculate the value of $\frac{1}{\sqrt[3]{7 + \pi}}$ to 3 significant figures. *(1 mark)*

10. Functions, 2ADV F1 2004 HSC 1c

Solve $\frac{x - 5}{3} - \frac{x + 1}{4} = 5$. *(2 marks)*

11. Functions, 2ADV F1 2005 HSC 1d

Express $\frac{(2x - 3)}{2} - \frac{(x - 1)}{5}$ as a single fraction in its simplest form. *(2 marks)*

12. Functions, 2ADV F1 2011 HSC 1a

Evaluate $\sqrt[3]{\frac{651}{4\pi}}$ correct to four significant figures. *(2 marks)*

13. Functions, 2ADV F1 2013 HSC 11a

Evaluate $\ln 3$ correct to three significant figures. *(1 mark)*

14. Functions, 2ADV F1 2008 HSC 1a

Evaluate $2\cos\left(\frac{\pi}{5}\right)$ correct to three significant figures. *(2 marks)*

15. Functions, 2ADV F1 2008 HSC 1e

Expand and simplify $(\sqrt{3} - 1)(2\sqrt{3} + 5)$. *(2 marks)*

16. Functions, 2ADV F1 2021 HSC 11

Solve $x + \frac{x - 1}{2} = 9$. *(2 marks)*

17. Functions, 2ADV F1 SM-Bank 46

Find a and b such that a, b are real numbers and

$$\frac{8 - \sqrt{27}}{2\sqrt{3}} = a + b\sqrt{3}. \quad (2 \text{ marks})$$

18. Functions, 2ADV F1 SM-Bank 47

Find a and b such that a, b are real numbers and

$$\frac{\sqrt{32} - 6}{3\sqrt{2}} = a + b\sqrt{2} \quad (2 \text{ marks})$$

19. Functions, 2ADV F1 SM-Bank 51

Find a and b such that a, b are real numbers and

$$\frac{\sqrt{3} - 2}{2\sqrt{3}} = a + b\sqrt{3}. \quad (2 \text{ marks})$$

20. Functions, 2ADV F1 SM-Bank 52

Find a and b such that a, b are real numbers and

$$\frac{6\sqrt{3} - \sqrt{5}}{2\sqrt{5}} = a + b\sqrt{15}. \quad (2 \text{ marks})$$

21. Functions, 2ADV F1 SM-Bank 54

Find the reciprocal of $\frac{1}{a} + \frac{1}{b} - \frac{c}{ab}$. *(2 marks)*

22. Functions, 2ADV F1 SM-Bank 56

Simplify $\frac{9x^2}{x + 3} \div \frac{3x}{x^2 - 9}$. *(2 marks)*

23. Functions, 2ADV F1 SM-Bank 55

Simplify $\frac{4p - 12p^2}{3} \times \frac{6p}{3p^2 - p}$. *(2 marks)*

24. Functions, 2ADV F1 EQ-Bank 22

- Worker A picks a bucket of blueberries in a hours. Worker B picks a bucket of blueberries in b hours.
- Write an algebraic expression for the fraction of a bucket of blueberries that could be picked in one hour if A and B worked together. **(2 marks)**
 - What does the reciprocal of this fraction represent? **(1 mark)**

25. Functions, 2ADV F1 2004 HSC 1a

The radius of Mars is approximately 3 397 000 metres. Write this number in scientific notation, correct to two significant figures. **(2 marks)**

26. Functions, 2ADV F1 2004 HSC 1d

Find integers a and b by showing working to expand and simplify

$$(3 - \sqrt{2})^2 = a - b\sqrt{2}. \text{ (2 marks)}$$

27. Functions, 2ADV F1 2008 HSC 1c

Simplify $\frac{2}{n} - \frac{1}{n+1}$. **(2 marks)**

28. Functions, 2ADV F1 EQ-Bank 10

Fully simplify the expression $\frac{4}{x^2 - 9} - \frac{2x + 1}{x + 3}$ **(3 marks)**

29. Functions, 2ADV F1 EQ-Bank 21

Simplify $\left(\frac{p}{q}\right)^3 \div (pq^{-2})$. **(2 marks)**

Worked Solutions

1. Functions, 2ADV F1 2018 HSC 1 MC

$$\begin{aligned} 7^{-1.3} &= 0.0796\dots \\ &= 0.08 \text{ (2 d.p.)} \\ \Rightarrow B \end{aligned}$$

2. Functions, 2ADV F1 2012 HSC 1 MC

$$\begin{aligned} 4.10 \\ \Rightarrow B \end{aligned}$$

3. Functions, 2ADV F1 2014 HSC 1 MC

$$\begin{aligned} \frac{\pi^2}{6} &= 1.6449\dots \\ &= 1.64 \text{ (3 sig. figures)} \\ \Rightarrow A \end{aligned}$$

4. Functions, 2ADV F1 2015 HSC 1 MC

$$\begin{aligned} 0.005 \ 233 \ 59 \\ &= 5.234 \times 10^{-3} \\ \Rightarrow D \end{aligned}$$

5. Functions, 2ADV F1 2019 HSC 1 MC

$$\begin{aligned} \pi^{10} &= 93648.04\dots \\ &= 9.4 \times 10^4 \\ \Rightarrow B \end{aligned}$$

6. Functions, 2ADV F1 2005 HSC 1a

$$\begin{aligned} \sqrt{\frac{275.4}{5.2 \times 3.9}} &= 3.685\dots \\ &= 3.7 \text{ (2 sig. figures)} \end{aligned}$$

7. Functions, 2ADV F1 2007 HSC 1a

$$\begin{aligned} \sqrt{\pi^2 + 5} &= 3.8561\dots \\ &= 3.86 \text{ (to 2 d.p.)} \end{aligned}$$

8. Functions, 2ADV F1 SM-Bank 50

$$\frac{1}{4\sqrt{3}} \times \frac{4\sqrt{3}}{4\sqrt{3}} = \frac{4\sqrt{3}}{16 \times 3}$$
$$= \frac{\sqrt{3}}{12}$$

9. Functions, 2ADV F1 SM-Bank 53

i. $\frac{1}{\sqrt[3]{7+\pi}} = (7+\pi)^{-\frac{1}{3}}$

ii. $\frac{1}{\sqrt[3]{7+\pi}} = 0.4619\dots$

$= 0.462 \text{ (to 3 sig. fig.)}$

10. Functions, 2ADV F1 2004 HSC 1c

$$\frac{x-5}{3} - \frac{x+1}{4} = 5$$
$$12\left(\frac{x-5}{3}\right) - 12\left(\frac{x+1}{4}\right) = 12 \times 5$$
$$4x - 20 - 3x - 3 = 60$$
$$x - 23 = 60$$
$$\therefore x = 83$$

11. Functions, 2ADV F1 2005 HSC 1d

$$\frac{(2x-3)}{2} - \frac{(x-1)}{5}$$
$$= \frac{5(2x-3) - 2(x-1)}{10}$$
$$= \frac{10x - 15 - 2x + 2}{10}$$
$$= \frac{8x - 13}{10}$$

12. Functions, 2ADV F1 2011 HSC 1a

$$\sqrt[3]{\frac{651}{4\pi}} = 3.72783\dots$$

$= 3.728 \text{ (to 4 sig. figures)}$

MARKER'S COMMENT: Show answer to 5 or 6 decimals before rounding. Correct rounding of a wrong answer still receives half marks.

13. Functions, 2ADV F1 2013 HSC 11a

$$\ln 3 = 1.09861\dots$$

$= 1.10 \text{ (to 3 sig. figures)}$

14. Functions, 2ADV F1 2008 HSC 1a

$$2\cos\left(\frac{\pi}{5}\right) = 1.6180\dots$$

$= 1.62 \text{ (3 sig)}$

15. Functions, 2ADV F1 2008 HSC 1e

$$(\sqrt{3}-1)(2\sqrt{3}+5)$$
$$= 2 \times 3 + 5\sqrt{3} - 2\sqrt{3} - 5$$
$$= 1 + 3\sqrt{3}$$

16. Functions, 2ADV F1 2021 HSC 11

$$x + \frac{x-1}{2} = 9$$
$$2x + x - 1 = 18$$
$$3x = 19$$
$$x = \frac{19}{3}$$

17. Functions, 2ADV F1 SM-Bank 46

$$\frac{8-\sqrt{27}}{2\sqrt{3}} \times \frac{2\sqrt{3}}{2\sqrt{3}} = \frac{2\sqrt{3}(8-3\sqrt{3})}{(2\sqrt{3})^2}$$
$$= \frac{16\sqrt{3}-18}{12}$$
$$= -\frac{3}{2} + \frac{4}{3}\sqrt{3}$$

$$\therefore a = -\frac{3}{2}, b = \frac{4}{3}$$

18. Functions, 2ADV F1 SM-Bank 47

$$\begin{aligned}\frac{\sqrt{32}-6}{3\sqrt{2}} \times \frac{3\sqrt{2}}{3\sqrt{2}} &= \frac{3\sqrt{2}(4\sqrt{2}-6)}{18} \\ &= \frac{24-18\sqrt{2}}{18} \\ &= \frac{4}{3} - \sqrt{2}\end{aligned}$$

$$\therefore a = \frac{4}{3}, b = -1$$

19. Functions, 2ADV F1 SM-Bank 51

$$\begin{aligned}\frac{\sqrt{3}-2}{2\sqrt{3}} &= \frac{\sqrt{3}-2}{2\sqrt{3}} \times \frac{2\sqrt{3}}{2\sqrt{3}} \\ &= \frac{2\sqrt{3}(\sqrt{3}-2)}{4 \times 3} \\ &= \frac{6-4\sqrt{3}}{12} \\ &= \frac{1}{2} - \frac{1}{3}\sqrt{3}\end{aligned}$$

$$\therefore a = \frac{1}{2}, b = -\frac{1}{3}$$

20. Functions, 2ADV F1 SM-Bank 52

$$\begin{aligned}\frac{6\sqrt{3}-\sqrt{5}}{2\sqrt{5}} &= \frac{6\sqrt{3}-\sqrt{5}}{2\sqrt{5}} \times \frac{2\sqrt{5}}{2\sqrt{5}} \\ &= \frac{2\sqrt{5}(6\sqrt{3}-\sqrt{5})}{4 \times 5} \\ &= \frac{12\sqrt{15}-10}{20} \\ &= -\frac{1}{2} + \frac{3}{5}\sqrt{15}\end{aligned}$$

$$\therefore a = -\frac{1}{2}, b = \frac{3}{5}$$

21. Functions, 2ADV F1 SM-Bank 54

$$\begin{aligned}\frac{1}{a} + \frac{1}{b} - \frac{c}{ab} &= \frac{b}{ab} + \frac{a}{ab} - \frac{c}{ab} \\ &= \frac{b+a-c}{ab}\end{aligned}$$

Reciprocal of $x = x^{-1}$

$$\therefore \text{Reciprocal of } \frac{b+a-c}{ab} = \left(\frac{b+a-c}{ab}\right)^{-1} = \frac{ab}{a+b-c}$$

22. Functions, 2ADV F1 SM-Bank 56

$$\begin{aligned}\frac{9x^2}{x+3} \div \frac{3x}{x^2-9} &= \frac{9x^2}{x+3} \times \frac{x^2-9}{3x} \\ &= \frac{9x^2}{x+3} \times \frac{(x-3)(x+3)}{3x} \\ &= 3x(x-3)\end{aligned}$$

23. Functions, 2ADV F1 SM-Bank 55

$$\begin{aligned}\frac{4p-12p^2}{3} \times \frac{6p}{3p^2-p} &= \frac{4p(1-3p)}{3} \times \frac{6p}{p(3p-1)} \\ &= \frac{8p(1-3p)}{3p-1} \\ &= -8p\end{aligned}$$

24. Functions, 2ADV F1 EQ-Bank 22

i. In one hour:

Worker A picks $\frac{1}{a}$ bucket.

Worker B picks $\frac{1}{b}$ bucket.

$\therefore$ Fraction picked in 1 hour working together

$$\begin{aligned}&= \frac{1}{a} + \frac{1}{b} \\ &= \frac{a+b}{ab}\end{aligned}$$

ii. The reciprocal represents the number of hours it would take to fill one bucket, with A and B working together.

COMMENT: Note that the question asks for "a fraction".

25. Functions, 2ADV F1 2004 HSC 1a

$$\begin{aligned} &3\,397\,000 \\ &= 3.397 \times 10^6 \\ &= 3.4 \times 10^6 \text{ m} \quad (2 \text{ sig figures}) \end{aligned}$$

26. Functions, 2ADV F1 2004 HSC 1d

$$\begin{aligned} (3 - \sqrt{2})^2 &= 9 - 6\sqrt{2} + (\sqrt{2})^2 \\ &= 9 - 6\sqrt{2} + 2 \\ &= 11 - 6\sqrt{2} \end{aligned}$$

$$\therefore a = 11, \quad b = 6$$

27. Functions, 2ADV F1 2008 HSC 1c

$$\begin{aligned} \frac{2}{n} - \frac{1}{n+1} &= \frac{2(n+1) - 1(n)}{n(n+1)} \\ &= \frac{2n+2-n}{n(n+1)} \\ &= \frac{n+2}{n(n+1)} \end{aligned}$$

28. Functions, 2ADV F1 EQ-Bank 10

$$\begin{aligned} \frac{4}{x^2-9} - \frac{2x+1}{x+3} &= \frac{4 - (2x+1)(x-3)}{(x+3)(x-3)} \\ &= \frac{4 - (2x^2 - 5x - 3)}{(x+3)(x-3)} \\ &= \frac{-(2x^2 - 5x - 7)}{(x+3)(x-3)} \\ &= \frac{-(2x-7)(x+1)}{(x^2-9)} \end{aligned}$$

29. Functions, 2ADV F1 EQ-Bank 21

$$\begin{aligned} \left(\frac{p}{q}\right)^3 \div (pq^{-2}) &= \frac{p^3}{q^3} \div \frac{p}{q^2} \\ &= \frac{p^3}{q^3} \times \frac{q^2}{p} \\ &= \frac{p^2}{q} \end{aligned}$$